

Statistical Guarantees in Data-Driven Nonlinear Control: Conformal Robustness for Stability and Safety

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Abstract—We present a true-dynamics-agnostic, statistically rigorous framework for establishing exponential stability and safety guarantees of closed-loop, data-driven nonlinear control. Central to our approach is the novel concept of *conformal robustness*, which robustifies the Lyapunov and zeroing barrier certificates of data-driven dynamical systems against model prediction uncertainties using conformal prediction. It quantifies these uncertainties by leveraging rank statistics of prediction scores over system trajectories, without assuming any specific underlying structure of the prediction model or distribution of the uncertainties. With the quantified uncertainty information, we further construct the conformally robust control Lyapunov function (CR-CLF) and control barrier function (CR-CBF), data-driven counterparts of the CLF and CBF, for fully data-driven control with statistical guarantees of finite-horizon exponential stability and safety. The performance of the proposed concept is validated in numerical simulations with four benchmark nonlinear control problems.

Index Terms—Data driven control, stability of nonlinear systems, uncertain systems.

I. INTRODUCTION

DATA-DRIVEN control is reshaping how autonomous systems operate under uncertainties, enabling decision-making without explicit knowledge of true dynamics. Yet, a formal approach to establishing their performance guarantees remains non-standardized, whereas existing methods often rely on (i) knowledge of underlying dynamics, (ii) structural constraints in prediction models, or (iii) assumptions on probability distributions or structures of system uncertainties.

Contributions: We propose *conformal robustness*, a statistically rigorous concept for robustifying finite-horizon stability and safety guarantees in data-driven nonlinear dynamical systems. Given any data-driven prediction models of the system dynamics, we quantify the model uncertainties over

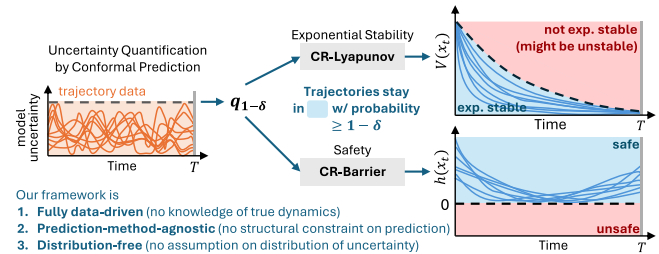


Fig. 1. Illustration of our proposed framework.

a finite time horizon using conformal prediction [1], [2], [3]. We then construct the conformally robust control Lyapunov function (CR-CLF) and control barrier function (CR-CBF) by robustifying the standard CLF and CBF conditions using the quantified uncertainty information. This enables systematic closed-loop control designs with provable finite-horizon stability and safety guarantees in data-driven nonlinear systems. Notably, unlike most existing methods, our proposed framework is fully data-driven and works without the need for (i) – (iii). Our proposed framework is illustrated in Fig. 1.

In simulations with four benchmark nonlinear control problems, we demonstrate the CR-CLFs and CR-CBFs in fully data-driven settings. We also show that they can be synthesized by quadratic programs and neural networks.

Related Work: Control Lyapunov functions (CLFs) [4], [5], [6] and control barrier functions (CBFs) [7], [8], [9] are among the most widely studied and applied tools for providing formal stability and safety guarantees in safety-critical nonlinear dynamical systems. They typically require partial knowledge of the true dynamics, and are not directly applicable to fully data-driven models with uncertainties. Most prior works assume specific underlying structures or probability distributions of the uncertainties. For instance, robust CLFs [6] and CBFs [10] require a known upper bound on the uncertainties. Stochastic CLFs [11] and CBFs [12] model the uncertainties using Wiener processes. Other approaches may impose specific structural assumptions on the uncertainties in the CLFs [13] and CBFs [14] and utilize machine learning to learn such uncertainties. Yet, the development of fully data-driven frameworks for constructing stability and safety guarantees that are robust against model uncertainties—with no structural or distributional assumptions—remains relatively unexplored.

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Data-driven control has also attracted significant attention from the statistical community. Some of their representative tools include, but are not limited to, conformal prediction [1], [2], [3], PAC-Bayes theory [15], scenario optimization [16] (see also the references therein). A thorough comparison among these methods in the context of data-driven controls can be found in [17]. This letter focuses on conformal prediction (CP), a statistical tool based on rank statistics for quantifying uncertainties in data-driven prediction models [1], [2], [3]. CP is well-suited to our problem setting because 1) it does not make any assumptions about the underlying distributions and structures of the uncertainties, 2) it is compatible with any prediction model, including neural networks [3], and 3) it requires only *exchangeability* [1], [2], [3], that is, the unseen real-world uncertainties are exchangeable with the empirically observed uncertainties. This assumption can be relaxed in some variations of CP [18], [19]. Consequently, CP-based methods have demonstrated successes in various domains of uncertainty-aware planning and control [19], [20], [21], [22], [23], [24]. For instance, CP has been used to obtain statistical coverage guarantees for data-driven dynamics models [22], data-driven state estimation [24], and environment prediction models [19], [20], to name a few, thereby constructing uncertainty-aware planning and safe control policies. In our proposed framework, we use CP to quantify model uncertainties over a finite time horizon, enabling its application in constructing stability and safety guarantees for continuous-time systems.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. Closed-Loop Guarantees for Nonlinear Systems

Consider a nonlinear dynamical system given by

$$\dot{x} = f(x, u), \quad (1)$$

where $x \in \mathcal{X} \subset \mathbb{R}^n$ is the state vector, $u \in \mathcal{U} \subset \mathbb{R}^m$ is the control input vector, and $f: \mathcal{X} \times \mathcal{U} \rightarrow \mathcal{X}$ is an unknown locally Lipschitz function.

Definition 1 (CLF [25]): A continuously differentiable (C^1) function $V: \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ is a control Lyapunov function (CLF) of (1) on $\mathcal{X}$ if $\exists c_1, c_2, c_3 \in \mathbb{R}_{>0}$ s.t. $\forall x \in \mathcal{X}$,

$$c_1 \|x\|^2 \leq V(x) \leq c_2 \|x\|^2 \quad (2)$$

and $\inf_{u \in \mathcal{U}} [\frac{\partial V}{\partial x} f(x, u)] \leq -c_3 V(x)$.

Lemma 1 [25]: The closed-loop system of (1) with any locally Lipschitz control policy $u(x) \in \{u \in \mathcal{U} \mid \frac{\partial V}{\partial x} f(x, u) \leq -c_3 V(x)\}$ is exponentially stable at $x = 0$. In other words, $V(x_t) \leq V(x_0)e^{-c_3 t}$ and $\|x_t\| \leq \sqrt{c_2/c_1} \|x_0\| e^{-c_3 t/2} \forall x_0 \in \mathcal{D} \subseteq \mathcal{X}$, where $\mathcal{D}$ is a region of attraction.

Definition 2 (CBF [9]): Let a safe set $\mathcal{C} \subset \mathcal{X}$ be the 0-superlevel set of a C^1 function $h: \mathcal{X} \rightarrow \mathbb{R}$, i.e., $\mathcal{C} = \{x \in \mathcal{X} \mid h(x) \geq 0\}$. If there exists an extended class- $\mathcal{K}_\infty$ function¹ α such that $\forall x \in \mathcal{X}$, $\sup_{u \in \mathcal{U}} [\frac{\partial h}{\partial x} f(x, u)] \geq -\alpha(h(x))$, then h is a control barrier function (CBF) of (1) on $\mathcal{C}$. In this letter, we consider $\alpha(h(x)) = \gamma h(x)$, where $\gamma \in \mathbb{R}_{>0}$.

Lemma 2 [9]: Any locally Lipschitz control policy $u(x) \in \{u \in \mathcal{U} \mid \frac{\partial h}{\partial x} f(x, u) \geq -\gamma h(x)\}$ renders $\mathcal{C}$ forward invariant (i.e., $\forall x_0 \in \mathcal{C}$, $x_t \in \mathcal{C} \forall t \in [0, \tau_f(x_0)]$), where $[0, \tau_f(x_0)]$ is the maximal time interval where x_t is defined).

¹An extended class- $\mathcal{K}_\infty$ function is a continuous, strictly increasing function $\alpha: \mathbb{R} \rightarrow \mathbb{R}$ which satisfies $\alpha(0) = 0$.

B. Finite-Horizon Exponential Stability and Safety

Remark 1: Since (1) is locally Lipschitz, for each $x_0 \in \mathcal{X}$, there exists a maximal time interval $[0, \tau_f(x_0)]$ where a unique solution exists. We can select a finite time horizon $T \leq \inf_{x_0 \in \mathcal{X}} \tau_f(x_0) \in \mathbb{R}_{>0}$ such that a unique solution to (1) exist on $[0, T] \forall x_0 \in \mathcal{X}$.

Extending the notion of finite-time stability in [26], we define finite-horizon exponential stability as follows.

Definition 3 (Finite-Horizon Exponential Stability): System (1) is finite-horizon exponentially stable at $x = 0$ in a finite time horizon $T \in \mathbb{R}_{>0}$ if there exists $U, W \in \mathbb{R}_{>0}$ such that $\|x_t\| \leq U \|x_0\| e^{-Wt} \forall t \in [0, T]$ and $\forall x_0 \in \mathcal{D}$.

Lemma 3: Suppose that a C^1 function $V: \mathcal{X} \rightarrow \mathbb{R}$ satisfies (2), and $V(x_t) \leq V(x_0)e^{-c_3 t} \forall t \in [0, T]$ and $\forall x_0 \in \mathcal{D}$. Then, system (1) is finite-horizon exponentially stable at $x = 0$ over a horizon $T \in \mathbb{R}_{>0}$. In particular, we have $\|x_t\| \leq \sqrt{c_2/c_1} \|x_0\| e^{-c_3 t/2} \forall t \in [0, T]$ and $\forall x_0 \in \mathcal{D}$.

We also define finite-horizon safety as follows.

Definition 4 (Finite-Horizon Safety): Given a safe set $\mathcal{C}$, system (1) is safe in a finite time horizon $T \in \mathbb{R}_{>0}$ if for all $x_0 \in \mathcal{C}$, the trajectories of (1) satisfy $x_t \in \mathcal{C} \forall t \in [0, T]$.

Lemma 4: Suppose that a safe set $\mathcal{C}$ is defined as in Definition 2, and there exists $\gamma \in \mathbb{R}_{>0}$ such that the C^1 function h satisfies $h(x_t) \geq h(x_0)e^{-\gamma t} \forall t \in [0, T]$ and $\forall x_0 \in \mathcal{C}$. Then, system (1) is safe in the time horizon T .

C. Split Conformal Prediction (Split CP)

Suppose we are given a dataset $S := \{(X_i, Y_i)\}_{i \in \mathcal{I}}$, where X_i and Y_i are the features and labels, and $\mathcal{I} = \{1, 2, \dots, N\}$ is a set of indices. First, we split the dataset S into two sets: a training set $S_{\text{train}} := \{(X_i, Y_i)\}_{i \in \mathcal{I}_{\text{train}}}$ for model learning and a calibration set $S_{\text{cal}} := \{(X_i, Y_i)\}_{i \in \mathcal{I}_{\text{cal}}}$ for uncertainty quantification. $\mathcal{I}_{\text{train}}$ and $\mathcal{I}_{\text{cal}}$ are sets of indices of S_{train} and S_{cal} , such that $\mathcal{I}_{\text{train}} \cap \mathcal{I}_{\text{cal}} = \emptyset$ and $\mathcal{I}_{\text{train}} \cup \mathcal{I}_{\text{cal}} = \mathcal{I}$. The sizes of the two sets are denoted by $N_{\text{train}} := |\mathcal{I}_{\text{train}}|$ and $N_{\text{cal}} := |\mathcal{I}_{\text{cal}}|$. Given any regression model $\hat{Y}_i = \hat{f}(X_i)$ trained on S_{train} , we define the nonconformity scores as $s_i := \|\hat{f}(X_i) - Y_i\| \forall i \in \mathcal{I}_{\text{cal}}$. Intuitively, smaller scores imply a more accurate regression model. Given a *failure probability* $\delta \in (0, 1)$, we define the conformal quantile as the $(1 - \delta)$ th quantile of the empirical distribution of $\{s_i\}_{i \in \mathcal{I}_{\text{cal}}} \cup \infty$ [3], [18], denoted by

$$q_{1-\delta} := \text{Quantile}\left(1 - \delta; \sum_{i=1}^{N_{\text{cal}}} \frac{1}{1+N_{\text{cal}}} \delta_{s_i} + \frac{1}{1+N_{\text{cal}}} \delta_\infty\right), \quad (3)$$

in which δ_a denotes the point mass probability at $a \in \mathbb{R} \cup \{-\infty, \infty\}$. This quantile can be equivalently interpreted as the $\lceil (1-\delta)(N_{\text{cal}}+1) \rceil$ -th smallest nonconformity score in $\{s_i\}_{i \in \mathcal{I}_{\text{cal}}}$. $\lceil \cdot \rceil$ denotes the ceiling function.

Lemma 5: [Split CP] Suppose we have a regression model $\hat{Y}_i = \hat{f}(X_i)$ trained on S_{train} . If, for any given new sample $(X_{\text{new}}, Y_{\text{new}})$, all the nonconformity scores in $\{s_i\}_{i \in \mathcal{I}_{\text{cal}} \cup \mathcal{I}_{\text{new}}}$ are exchangeable, then we have the following statistical bound on the prediction error:

$$\mathbb{P}\{\|\hat{f}(X_{\text{new}}) - Y_{\text{new}}\| \leq q_{1-\delta} \mid S_{\text{train}}\} \geq 1 - \delta.$$

Proof: See [18, Th. 1 and Section VI-A]. ■

The exchangeability condition in Lemma 5 means that the joint probability distributions of $\{s_i\}_{i \in \mathcal{I}_{\text{cal}} \cup \mathcal{I}_{\text{new}}}$ and $\{s_{\sigma(i)}\}_{i \in \mathcal{I}_{\text{cal}} \cup \mathcal{I}_{\text{new}}}$ are identical under any permutation σ .

Remark 2: The statistical coverage in Lemma 5 is affected by the choice of δ , but not N_{train} and N_{cal} . A smaller δ means a larger $q_{1-\delta}$ (see (3)), and vice versa.

Remark 3 [3]: If conditioned on both S_{train} and S_{cal} , the statistical coverage in Lemma 5 becomes $\mathbb{P}\{\|f(X_{i_{\text{new}}}) - Y_{i_{\text{new}}}\| \leq q_{1-\delta} \mid S_{\text{train}} \cup S_{\text{cal}}\} \sim \text{Beta}(k_\delta, N_{\text{cal}} + 1 - k_\delta)$, where $\text{Beta}(\cdot, \cdot)$ denotes a beta distribution and $k_\delta = \lceil (1 - \delta)(N_{\text{cal}} + 1) \rceil$. Note that as $N_{\text{cal}} \rightarrow \infty$, the mean of the beta distribution tends to $1 - \delta$ and the variance tends to 0.

To relax the exchangeability assumption, we could use weighted conformal prediction [18], [22].

Lemma 6 (Weighted Split CP [18]): First, assign custom weights $\{w_i\}_{i \in \mathcal{I}_{\text{cal}}}$, where $w_i \in [0, 1]$, to data in S_{cal} . Define the quantile as $q_{1-\delta}^w := \text{Quantile}(1 - \delta; \sum_{i=1}^{N_{\text{cal}}} \tilde{w}_i \delta_{s_i} + \tilde{w}_\infty \delta_\infty)$, where $\tilde{w}_i = w_i / (\sum_{j=1}^{N_{\text{cal}}} w_j + 1)$ and $\tilde{w}_\infty = 1 / (\sum_{j=1}^{N_{\text{cal}}} w_j + 1)$. Then, the statistical coverage becomes $\mathbb{P}\{\|\hat{f}(X_{i_{\text{new}}}) - Y_{i_{\text{new}}}\| \leq q_{1-\delta}^w \mid S_{\text{train}}\} \geq 1 - \delta - \sum_{i=1}^{N_{\text{cal}}} \tilde{w}_i \cdot d_{\text{TV}}(Z, Z^i)$, where $d_{\text{TV}}(\cdot, \cdot)$ is the total variation distance between two distributions, $Z = \{s_i\}_{i \in \mathcal{I}_{\text{cal}} \cup i_{\text{new}}}$, and Z^i is the score vector with s_i swapped with $s_{i_{\text{new}}}$ in Z .

The weights $\{w_i\}_{i \in \mathcal{I}_{\text{cal}}}$ are chosen to minimize the coverage gap $\sum_{i=1}^{N_{\text{cal}}} \tilde{w}_i \cdot d_{\text{TV}}(Z, Z^i)$, with higher weights assigned to data points similar in distribution to the new sample [18]. See [18], [22] for details.

D. Problem Formulation

In this letter, we consider the case where the true system (1) is unknown. Suppose we are given a dataset of trajectories sampled from (1). The dataset is denoted by $S := \{\phi(t; x_0^i, u(\cdot), T), f(\phi(t; x_0^i, u(\cdot), T), u(\cdot))\}_{i \in \mathcal{I}}$, in which $\phi(t; x_0^i, u(\cdot), T)$ denotes the trajectory that solves (1) in $t \in [0, T]$ with initial condition x_0^i ; $u(\cdot) \in \mathcal{U}$ is any control input that renders $f(x, u(\cdot))$ locally Lipschitz; $T \in \mathbb{R}_{>0}$ is a finite time horizon. For the simplicity of notation, we denote $\phi_t(x_0^i) := \phi(t; x_0^i, u(\cdot), T)$ in the rest of this letter. Additionally, suppose we are given a data-driven model $\hat{f}: \mathcal{X} \times \mathcal{U} \rightarrow \mathcal{X}$ that is learned from S . We require that $\hat{f}(x, u(\cdot))$ is locally Lipschitz.

Without knowing the true system (1), we can only define the guarantees based on the learned model $\hat{f}$. However, no matter how accurate $\hat{f}$ is, the guarantees defined using $\hat{f}$ can still fail when applied to the true system. This letter aims to address this issue by solving the following problem.

Problem 1: Suppose we are given a data-driven model $\dot{x} = \hat{f}(x, u)$ with a level of uncertainty described by $\mathbb{P}\{\sup_{t \in [0, T]} \|f(\phi_t(x_0), u(\cdot)) - \hat{f}(\phi_t(x_0), u(\cdot))\| \leq q(\delta)\} \geq 1 - \delta$, where x_0 is any initial state in $\mathcal{X}$, $\delta \in (0, 1)$ is the failure probability, and $q(\delta)$ is an upper bound that is a function of δ . Our goal is to construct the guarantees using $\hat{f}$ in conjunction with $q(\delta)$, such that the true system (1) is provably exponentially stable and safe within a time horizon T with high probability.

III. CONFORMAL ROBUSTNESS

In this section, we will address Problem 1 by incorporating the uncertainty quantification method of Section III-A into the formal guarantees of Sections II-A and II-B.

A. Uncertainty Quantification With Conformal Prediction

To solve Problem 1, we first utilize split CP to quantify the uncertainty of the data-driven model. We split the dataset S into a training set $S_{\text{train}} := \{\phi_t(x_0^i), f(\phi_t(x_0^i), u(\cdot))\}_{i \in \mathcal{I}_{\text{train}}}$ and a calibration set $S_{\text{cal}} := \{\phi_t(x_0^i), f(\phi_t(x_0^i), u(\cdot))\}_{i \in \mathcal{I}_{\text{cal}}}$. Given a data-driven model $\hat{f}$ learned from S_{train} , we define the non-conformity scores using S_{cal} as $s_i := \sup_{t \in [0, T]} \|\Delta(\phi_t(x_0^i), u(\cdot))\| \forall i \in \mathcal{I}_{\text{cal}}$, where $\Delta(\cdot, \cdot) := f(\cdot, \cdot) - \hat{f}(\cdot, \cdot)$ denotes the model uncertainty. To utilize Lemma 5, we make the following assumption.

Assumption 1: For any newly given sample $(\phi_t(x_0^{i_{\text{new}}}), f(\phi_t(x_0^{i_{\text{new}}}), u(\cdot)))$, all the nonconformity scores in $\{s_i\}_{i \in \mathcal{I}_{\text{cal}} \cup i_{\text{new}}}$ are exchangeable. We do not assume $u(\cdot)$ to be the same across all samples. We only assume $u(\cdot)$ renders $f(x, u(\cdot))$ locally Lipschitz continuous.

Remark 4: To relax the exchangeability assumption, we could adopt weighted split CP (Lemma 6). This will introduce a coverage gap in the statistical coverage while the rest of our proposed theorems (to be presented in Sections III-B and III-C) remain unchanged. If the real-world distribution shifts online, we could update δ of (3) using adaptive CP [19] also with minor modifications of our proposed theorems. Hence, for clarity of presentation and due to page limitations, this letter focuses on the standard split CP in Lemma 5, which can be fundamental for these non-exchangeable CP variations, with rigorous arguments left for future work.

The following corollary serves as our uncertainty quantification metric for the data-driven model.

Corollary 1: Suppose we are given a data-driven model $\hat{f}$ learned from S_{train} . If Assumption 1 holds for any $x_0 = x_0^{i_{\text{new}}} \in \mathcal{X}$, then by Lemma 5, $\mathbb{P}\{\sup_{t \in [0, T]} \|\Delta(\phi_t(x_0), u(\cdot))\| \leq q_{1-\delta}\} \geq 1 - \delta$. This also implies $\mathbb{P}\{\sup_{s \in [0, t]} \|\Delta(\phi_s(x_0), u(\cdot))\| \leq q_{1-\delta}, \forall t \in [0, T]\} \geq 1 - \delta$.

Remark 5: We consider the case where S sampled from (1) is noise-free. However, if measurement noise is present, the CP-based method in [24] can still be incorporated into our framework to quantify the state estimation uncertainty.

B. Conformally Robust Control Lyapunov Function

Definition 5 (CR-CLF): Suppose we are given a data-driven model $\hat{f}$ and a conformal quantile $q_{1-\delta}$ defined in (3). A C^1 function $V: \mathcal{X} \rightarrow \mathbb{R}$ is a conformally robust CLF (CR-CLF) if it satisfies (2), and $\forall x \in \mathcal{X}, \exists u \in \mathcal{U}$ s.t.

$$\frac{\partial V}{\partial x} \hat{f}(x, u) + c_3 V(x) + \left\| \frac{\partial V}{\partial x} \right\| \cdot q_{1-\delta} \leq 0. \quad (4)$$

Theorem 1: Consider a set of control inputs given by $\mathbf{K}_{\text{CL}}(x) = \{u \in \mathcal{U} \mid \frac{\partial V}{\partial x} \hat{f}(x, u) + c_3 V(x) + \left\| \frac{\partial V}{\partial x} \right\| q_{1-\delta} \leq 0\}$. Any locally Lipschitz continuous control policy $u(x) \in \mathbf{K}_{\text{CL}}(x)$ renders $\mathbb{P}\{V(x_t) \leq V(x_0)e^{-c_3 t}, \forall t \in [0, T]\} \geq 1 - \delta$ and $\mathbb{P}\{\|x_t\| \leq \sqrt{c_2/c_1} \|x_0\| e^{-c_3 t/2}, \forall t \in [0, T]\} \geq 1 - \delta$, for all $x_0 \in \mathcal{D}$.

Proof: Since $\frac{\partial V}{\partial x} \hat{f}(x, u) + c_3 V(x) + \left\| \frac{\partial V}{\partial x} \right\| q_{1-\delta} \leq 0$, we have $\dot{V} = \frac{\partial V}{\partial x} (\hat{f}(x, u) + \Delta(x, u)) \leq -c_3 V(x) + \left\| \frac{\partial V}{\partial x} \right\| (\|\Delta(x, u)\| - q_{1-\delta})$. Integrating the inequality over time interval $[0, t]$, where $t \in [0, T]$, yields $V(x_t) \leq V(x_0)e^{-c_3 t} + (\sup_{s \in [0, t]} \|\Delta(\phi_s(x_0), u(\cdot))\| - q_{1-\delta}) \cdot \int_0^t e^{-c_3(t-\tau)} \left\| \frac{\partial V}{\partial x} \right\| d\tau$, where $x_0 \in \mathcal{D}$. By Corollary 1, $\mathbb{P}\{V(x_t) \leq V(x_0)e^{-c_3 t}, \forall t \in [0, T]\} \geq 1 - \delta$. By Lemma 3, $\mathbb{P}\{\|x_t\| \leq \sqrt{c_2/c_1} \|x_0\| e^{-c_3 t/2}, \forall t \in [0, T]\} \geq 1 - \delta$. ■

Sometimes, we are not able to construct a CR-CLF that exactly satisfies (4). Instead, our approximation of the CR-CLF might induce small violations of (4). The following proposition analyzes the robustness of the CR-CLF against such violations.

Proposition 1 (Approximated CR-CLF): Suppose we are given $\hat{f}$ and $q_{1-\delta}$ as in Definition 5. A C^1 function $\hat{V} : \mathcal{X} \rightarrow \mathbb{R}$ is an approximated CR-CLF if it satisfies (2), and $\forall x \in \mathcal{X}, \exists u \in \mathcal{U}$ s.t.

$$\frac{\partial \hat{V}}{\partial x} \hat{f}(x, u) + c_3 \hat{V}(x) + \left\| \frac{\partial \hat{V}}{\partial x} \right\| \cdot q_{1-\delta} \leq r(x),$$

where $r(x) \in \mathbb{R}_{\geq 0}$ is the violation. Consider $\mathbf{K}_{\text{CL}}^r(x) = \{u \in \mathcal{U} \mid \frac{\partial \hat{V}}{\partial x} \hat{f}(x, u) + c_3 \hat{V}(x) + \left\| \frac{\partial \hat{V}}{\partial x} \right\| q_{1-\delta} \leq r(x)\}$. If we implement a locally Lipschitz continuous control policy $u(x) \in \mathbf{K}_{\text{CL}}^r(x)$, then under the exchangeability assumption, we can apply split CP on the violations over trajectories and obtain $\mathbb{P}\{\sup_{s \in [0, t]} r(\phi_s(x_0)) \leq \bar{r}, \forall t \in [0, T]\} \geq 1 - \delta_r$, where $\bar{r}$ is the conformal quantile. We further assume that the conformal coverage of the violations is independent of that of the model uncertainties. Therefore, we have $\mathbb{P}\{\hat{V}(x_t) \leq \hat{V}(x_0)e^{-c_3 t} + \bar{r}(1 - e^{-c_3 t})/c_3, \forall t \in [0, T]\} \geq (1 - \delta)(1 - \delta_r)$ and $\mathbb{P}\{\|x_t\| \leq \sqrt{(c_2/c_1)\|x_0\|^2 e^{-c_3 t} + \bar{r}(1 - e^{-c_3 t})/(c_1 c_3)}, \forall t \in [0, T]\} \geq (1 - \delta)(1 - \delta_r)$, where $x_0 \in \mathcal{D}$.

C. Conformally Robust Control Barrier Function

Definition 6 (CR-CBF): Suppose we are given a data-driven model $\hat{f}$ and a conformal quantile $q_{1-\delta}$ defined in (3). A C^1 function $h : \mathcal{X} \rightarrow \mathbb{R}$ is a conformally robust control barrier function (CR-CBF) if the safe set $\mathcal{C}$ is its 0-superlevel set, and $\forall x \in \mathcal{X}, \exists u \in \mathcal{U}$ s.t.

$$\frac{\partial h}{\partial x} \hat{f}(x, u) + \gamma h(x) - \left\| \frac{\partial h}{\partial x} \right\| \cdot q_{1-\delta} \geq 0. \quad (5)$$

Theorem 2: Consider a set of control inputs given by $\mathbf{K}_{\text{CB}}(x) = \{u \in \mathcal{U} \mid \frac{\partial h}{\partial x} \hat{f}(x, u) + \gamma h(x) - \left\| \frac{\partial h}{\partial x} \right\| q_{1-\delta} \geq 0\}$. Any locally Lipschitz continuous control policy $u(x) \in \mathbf{K}_{\text{CB}}(x)$ renders $\mathbb{P}\{x_t \in \mathcal{C}, \forall t \in [0, T] \mid x_0 \in \mathcal{C}\} \geq 1 - \delta$.

Proof: Since $\frac{\partial h}{\partial x} \hat{f}(x, u) + \gamma h(x) - \left\| \frac{\partial h}{\partial x} \right\| q_{1-\delta} \geq 0$, we have $\dot{h} = \frac{\partial h}{\partial x} (\hat{f}(x, u) + \Delta(x, u)) \geq -\gamma h(x) + \left\| \frac{\partial h}{\partial x} \right\| (-\|\Delta(x, u)\| + q_{1-\delta})$. Integrating the inequality above over $[0, t]$, where $t \in [0, T]$, with any initial state $x_0 \in \mathcal{C}$ yields $h(x_t) \geq h(x_0)e^{-\gamma t} - (\sup_{s \in [0, t]} \|\Delta(\phi_s(x_0), u(\cdot))\| - q_{1-\delta}) \cdot \int_0^t e^{-\gamma(t-\tau)} \left\| \frac{\partial h}{\partial x} \right\| d\tau$. By Corollary 1, we have $\mathbb{P}\{h(x_t) \geq h(x_0)e^{-\gamma t}, \forall t \in [0, T]\} \geq 1 - \delta$. Therefore, by Lemma 4, $\mathbb{P}\{x_t \in \mathcal{C}, \forall t \in [0, T] \mid x_0 \in \mathcal{C}\} \geq 1 - \delta$. ■

Sometimes, we can only approximate the CR-CBF with violations of (5). The robustness of the CR-CBF against such violations is discussed in the following proposition.

Proposition 2 (Approximated CR-CBF): Suppose we are given $\hat{f}$ and $q_{1-\delta}$ as in Definition 6. A C^1 function $\hat{h} : \mathcal{X} \rightarrow \mathbb{R}$ is an approximated CR-CBF if the safe set $\mathcal{C}$ is its 0-superlevel set, and $\forall x \in \mathcal{X}, \exists u \in \mathcal{U}$ s.t.

$$\frac{\partial \hat{h}}{\partial x} \hat{f}(x, u) + \gamma \hat{h}(x) - \left\| \frac{\partial \hat{h}}{\partial x} \right\| \cdot q_{1-\delta} \geq -r(x),$$

where $r(x) \in \mathbb{R}_{\geq 0}$ is the violation. Consider $\mathbf{K}_{\text{CB}}^r(x) = \{u \in \mathcal{U} \mid \frac{\partial \hat{h}}{\partial x} \hat{f}(x, u) + \gamma \hat{h}(x) - \left\| \frac{\partial \hat{h}}{\partial x} \right\| q_{1-\delta} \geq -r(x)\}$. Any locally Lipschitz continuous control policy $u(x) \in \mathbf{K}_{\text{CB}}^r(x)$ renders $\mathbb{P}\{\hat{h}(x_t) \geq \hat{h}(x_0)e^{-\gamma t} - \bar{r}(1 - e^{-\gamma t})/\gamma, \forall t \in [0, T] \mid x_0 \in \mathcal{C}\} \geq (1 - \delta)(1 - \delta_r)$, where $\bar{r}$ and δ_r are obtained by split CP as described in Proposition 1.

IV. CASE STUDIES

We consider four benchmark nonlinear systems. Using only data and without knowing the true dynamics of these systems, we implement the CR-CLF and CR-CBF to validate Theorems 1 and 2.

In this section, we will use the sparse identification of nonlinear dynamics (SINDy) algorithm [27] for model learning. The data-driven model is parameterized by $\dot{x} = \hat{f}(x, u) = \xi \Theta^\top(x, u)$, where $\Theta^\top : \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}^{M \times 1}$ is a regressor, consisting of M locally Lipschitz continuous basis functions; $\xi \in \mathbb{R}^{n \times M}$ is a parameter matrix to be learned, of which each row is a 1-by- M parameter vector. Given a training dataset S_{train} , ξ is learned using sparse regression [27]. If the learned model is control-affine, let it also be expressed as $\dot{x} = \hat{f}(x, u) = \hat{a}(x) + \hat{b}(x)u$.

A. Example 1: Inverted Pendulum With CR-CLF

We utilize the CR-CLF to stabilize an inverted pendulum. Let the state $x = [\theta \ \dot{\theta}]^\top \in \mathbb{R}^2$, where θ denotes the angular position of the pendulum, and control $u \in \mathbb{R}$ be the torque input. The true dynamics of the inverted pendulum is $\dot{x} = \begin{bmatrix} \dot{\theta} \\ -b\dot{\theta}/I + mgL \sin \theta / (2I) \end{bmatrix} + \begin{bmatrix} 0 \\ -1/I \end{bmatrix} u$, where $m = 1$ kg and $L = 1$ m are the mass and length of the pendulum, $b = 0.01$ is the damping, and $I = mL^2/3$ is the moment of inertia. For model learning, we choose $\Theta(x, u) = [1 \ \theta \ \dot{\theta} \ \theta^2 \ \theta \dot{\theta} \ \dot{\theta}^2, \dots, \theta^5 \ \theta^4 \dot{\theta} \ \theta^3 \dot{\theta}^2 \ \dots, \dot{\theta}^5] \otimes [1 \ u]$. Applying CP gives $q_{1-\delta} = 0.196$ for $T = 5$ s and $\delta = 0.1$.

We construct the CLF candidate as $V = x^\top P x$, where P solves $A^\top P + PA + Q = 0$. A is the Jacobian of the closed-loop system of $\hat{f}$ with feedback control $u = [8 \ 5]x$ and $Q = c_3 I_{2 \times 2}$ [28], where $c_3 = 0.5$ is the exponential decay rate. By construction, such V satisfies (2). Since $\hat{f}(x, u)$ is control-affine, we can synthesize a control policy that satisfies (4) by solving the following quadratic program (QP) [9]:

$$u_{\text{CR-CLF}}^*(x) = \arg \min_{u \in \mathbf{K}_{\text{CL}}(x)} \|u\|^2. \quad (\text{CR-CLF QP})$$

For comparison, we also implement a regular, uncertainty-agnostic CLF QP controller $u_{\text{CLF}}^*(x) = \arg \min_{u \in \mathbf{K}_{\text{L}}(x)} \|u\|^2$, where $\mathbf{K}_{\text{L}}(x) = \{u \in \mathcal{U} \mid \frac{\partial V}{\partial x} \hat{f}(x, u) + c_3 V(x) \leq 0\}$.

Proposition 3: If $\hat{a}(x)$, $\hat{b}(x)$, and $\frac{\partial V}{\partial x}(x)$ are locally Lipschitz continuous, and $\frac{\partial V}{\partial x} \hat{b}(x) \neq 0$, then $u_{\text{CR-CLF}}^*(x)$ is locally Lipschitz continuous (see [9] for proof), thus satisfying Theorem 1.

The code used for simulation is adapted from [28]. For each controller, we simulate 30 trajectories starting from 30 random initial states in $\{x \in \mathcal{X} \mid V(x) = 1.3\}$. Fig. 2 shows that $u_{\text{CR-CLF}}^*(x)$ given by (CR-CLF QP) renders finite-horizon exponential stability of the true system for $T = 5$ s, while $u_{\text{CLF}}^*(x)$ given by the regular CLF QP fails to achieve the same stability guarantee.

B. Example 2: Adaptive Cruise Control With CR-CBF

We utilize the CR-CBF to guarantee finite-horizon safety for an adaptive cruise control (ACC) system [8], which consists of a leader and a follower vehicle. The follower is supposed to track a desired velocity while maintaining a safe distance from the leader. Let the state $x = [p \ v \ z]^\top \in \mathbb{R}^3$, where p and v denote the position and velocity of the follower, and z denote the distance between the leader and the follower.

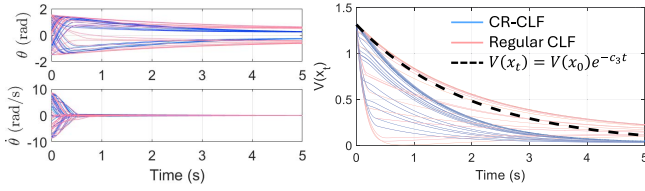


Fig. 2. Inverted pendulum: comparison between the CR-CLF and regular CLF.

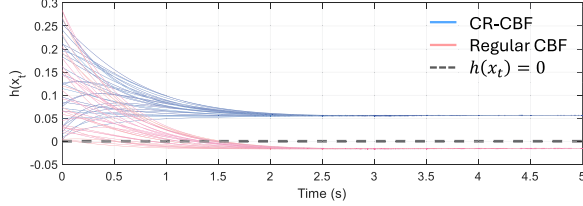


Fig. 3. Adaptive cruise control: comparison between the CR-CBF and regular CBF.

Let control input $u \in \mathbb{R}$ be the force acting on the follower.

The true system is given by $\dot{x} = \begin{bmatrix} v \\ -\frac{1}{m}(f_0 + f_1 v + f_2 v^2) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u$, where $v_0 = 15$ m/s is the velocity of the leader; $m = 2000$ kg is the mass of the follower; $f_0 = 0.5$, $f_1 = 5$, and $f_2 = 1$ are parameters [8]. For model learning, we choose $\Theta(x, u) = [1 \ p \ v \ z] \otimes [1 \ u]$. Applying CP gives $q_{1-\delta} = 0.102$ for $T = 5$ s and $\delta = 0.05$.

We define the safe set $\mathcal{C} = \{(p, v, z) \mid z - T_h v \geq 0\}$, where $T_h = 1$ s is the look-ahead time [8]. A straightforward choice of the CBF candidate is $h(x) = z - T_h v$. Since $\hat{f}(x, u)$ is control-affine, we can synthesize a control policy such that (5) is satisfied by solving the following QP [9]:

$$u_{\text{CR-CBF}}^*(x) = \arg \min_{u \in \mathbf{K}_{\text{CB}}(x)} \|u - \bar{u}(x)\|^2, \quad (\text{CR-CBF QP})$$

in which $\bar{u}(x)$ is a reference control policy. In this example, we set $\gamma = 2$ in $\mathbf{K}_{\text{CB}}(x)$ and $\bar{u} = 100(v_d - v)$, where $v_d = 20$ m/s is the desired velocity. For comparison, we also implement a regular, uncertainty-agnostic CBF QP controller by solving $u_{\text{CBF}}^*(x) = \arg \min_{u \in \mathbf{K}_{\text{CB}}(x)} \|u - \bar{u}(x)\|^2$, where $\mathbf{K}_{\text{CB}}(x) = \{u \in \mathcal{U} \mid \frac{\partial h}{\partial x} \hat{f}(x, u) + \gamma h(x) \geq 0\}$.

Proposition 4: If $\hat{a}(x)$, $\hat{b}(x)$, and $\frac{\partial h}{\partial x}(x)$ are locally Lipschitz continuous, and $\frac{\partial h}{\partial x} \hat{b}(x) \neq 0$, then $u_{\text{CR-CBF}}^*(x)$ is locally Lipschitz continuous (see [9] for proof), thus satisfying Theorem 2.

The code used for simulation is adapted from [28]. We simulate 30 trajectories starting from 30 different initial states in the safe set $\mathcal{C}$. Fig. 3 shows that $u_{\text{CR-CBF}}^*(x)$ renders finite-horizon safety for $T = 5$ s, while $u_{\text{CBF}}^*(x)$ renders unsafe behavior (i.e., $h(x_t) < 0$).

C. Example 3: Dubins Car With Neural CR-CBF

We utilize a CR-CBF to maneuver a Dubins car towards a target while avoiding an obstacle. Let the state $x = [p_x \ p_y \ \theta]^\top \in \mathbb{R}^3$, where p_x and p_y are the x and y positions and θ is the heading. Let control input $u \in \mathbb{R}$ be the yaw rate. The true model of the Dubins car is given by $\dot{x} = [v \cos \theta \ v \sin \theta \ 0]^\top + [0 \ 0 \ 1]^\top u$, where $v = 1$ m/s is a constant speed. For model learning, we choose a regressor $\Theta(x, u) = [1 \ \theta \ \theta^2 \ \theta^3] \otimes [1 \ u]$. Applying CP gives $q_{1-\delta} = 0.554$ for $T = 10$ s and $\delta = 0.1$.

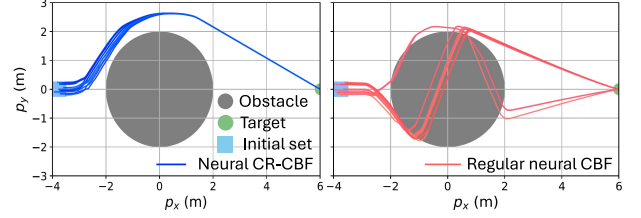


Fig. 4. Dubins car collision avoidance: comparison between the neural CR-CBF and regular neural CBF.

Consider a circular obstacle with a radius of 2 m centered at the origin and a target at (6, 0) m. Starting from the left of the obstacle, the goal is to reach the target while avoiding the obstacle. We defined the safe set as $\mathcal{C} = \{(p_x, p_y, \theta) \mid p_x^2 + p_y^2 > 2^2; |\theta| < \pi/2\}$. We adopt the framework in [29] to learn the CR-CBF using neural networks. The neural CR-CBF, denoted as $h_{\theta_B}(x)$, is parameterized by θ_B . The learning is formulated as an empirical loss minimization problem: $h_{\theta_B}^* = \min_{\theta_B} \mathcal{L}_B(h_{\theta_B})$. The loss is defined by $\mathcal{L}_B = \frac{\lambda_s}{N_s} \sum_{x_i \in \mathcal{C}} \text{ReLU}(-h_{\theta_B}(x_i)) + \frac{\lambda_u}{N_u} \sum_{x_i \in \mathcal{U}} \text{ReLU}(h_{\theta_B}(x_i)) + \frac{\lambda_b}{N} \sum_{x_i \in \mathcal{X}} r(x_i)$, where $N = 10^4$ denotes the total number of samples x_i uniformly sampled from $\mathcal{X}$; N_s (resp. N_u) denotes the number of samples in the safe (resp. unsafe) set; $\lambda_s = \lambda_u = 100$ and $\lambda_b = 1$ are the penalty weights; $r(x_i)$ is the slack variable obtained by solving $\min_{u \in \mathbf{K}_{\text{CB}}^r(x), r \in \mathbb{R}_{\geq 0}} \|u - \bar{u}(x)\|^2 + \lambda_r r$, where $\lambda_r = 10^4$ and $\bar{u}(x) = 10 \cdot \text{wrapToPi}(\text{atan2}(0 - p_y, 6 - p_x) - \theta)$. This QP becomes identical to (CR-CBF QP) when $r \equiv 0$. For comparison, we also learn a regular, uncertainty-agnostic neural CBF using the same method. For the regular neural CBF, $r(x_i)$ is obtained by solving $\min_{u \in \mathbf{K}_{\text{B}}^r(x), r \in \mathbb{R}_{\geq 0}} \|u - \bar{u}(x)\|^2 + \lambda_r r$, where $\mathbf{K}_{\text{B}}^r(x) = \{u \in \mathcal{U} \mid \frac{\partial h_{\theta_B}}{\partial x} \hat{f}(x, u) + \gamma h_{\theta_B}(x) \geq -r(x)\}$.

Fig. 4 shows that the neural CR-CBF QP controller drives the Dubins car towards the target while avoiding the obstacle in the 10-second time horizon. However, the regular neural CBF QP controller results in collisions with the obstacle.

D. Example 4: Cartpole With Neural CR-CLF

We utilize a CR-CLF to stabilize a cartpole at its upright position. Let the state $x = [z \ \theta \ \dot{z} \ \dot{\theta}]^\top \in \mathbb{R}^4$, where z is the position of the cart, and θ is the angular position of the pendulum. Let control input $u \in \mathbb{R}$ be the force acting on the cart. The true system is given by $\ddot{z} = \frac{u - m L \dot{\theta}^2 \sin(2\theta) + 0.5 m g \sin(2\theta)}{M + m \sin^2 \theta}$ and $\ddot{\theta} = \frac{u \cos \theta - 0.5 m L \dot{\theta}^2 \sin(2\theta) + (m + M) g \sin \theta}{(M + m \sin^2 \theta) L}$, where $M = 1$ kg and $m = 0.3$ kg are the mass of the cart and the pendulum, and $L = 1$ m is the length of the pendulum. For model learning, we choose $\Theta(x, u) = [1 \ \dot{z} \ \dot{\theta} \ z^2 \ \dot{z} \dot{\theta} \ \dot{\theta}^2] \otimes [1 \ \sin \theta \ \cos \theta] \otimes [1 \ \sin \theta \ \cos \theta] \otimes [1 \ \sin \theta \ \cos \theta] \otimes [1 \ \sin \theta \ \cos \theta] \otimes [1 \ u]$. Applying CP gives $q_{1-\delta} = 0.144$ for $T = 5$ s and $\delta = 0.05$.

We adapt the neural CLF in [30] to jointly learn the CR-CLF, $V_{\theta_V}(x)$, and its stabilizing controller, $u_{\theta_u}(x)$, using neural networks. θ_V and θ_u are the parameters to be learned. By construction in [30], V_{θ_V} satisfies condition (2) and $u_{\theta_u}(x)$ is locally Lipschitz continuous. The learning of the CR-CLF is formulated as an empirical loss minimization problem: $V_{\theta_V}^*, u_{\theta_u}^* = \min_{\theta_V, \theta_u} \mathcal{L}_L(V_{\theta_V}, u_{\theta_u})$. We consider $\mathcal{L}_L = \mathcal{L}_d + \mathcal{L}_a$, in which $\mathcal{L}_d = \frac{\lambda_d}{N} \sum_{i=1}^N \text{ReLU}(\frac{\partial V_{\theta_V}}{\partial x} \hat{f}(x_i, u_{\theta_u}) + c_3 V_{\theta_V}(x_i) + \|\frac{\partial V_{\theta_V}}{\partial x}\| q_{1-\delta})$ penalizes the violation of (4), and

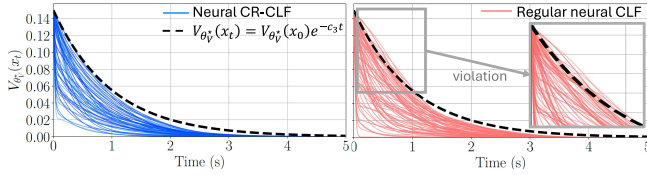


Fig. 5. Cartpole: comparison between the neural CR-CLF and regular neural CLF.

$\mathcal{L}_a = \frac{\lambda_a}{N} \sum_{i=1}^N \text{ReLU}(-V_{\theta_V}(x_i) + \beta \|x_i\|)$ encourages the growth of the region of attraction [31]. N denotes the number of samples. We set $\lambda_d = 500$, $c_3 = 1$, $\lambda_a = 300$, $\beta = 0.15$. For comparison, we also learn a regular, uncertainty-agnostic neural CLF using the same method with $\mathcal{L}_d = \frac{\lambda_d}{N} \sum_{i=1}^N \text{ReLU}(\frac{\partial V_{\theta_V}}{\partial x} \hat{f}(x_i, u_{\theta_u}) + c_3 V_{\theta_V}(x_i))$.

We simulate trajectories using both the neural CR-CLF and the regular neural CLF starting from 100 different initial states in $\{x \mid V_{\theta_V}(x) = 0.15\}$. Fig. 5 shows that the neural CR-CLF controller renders finite-horizon exponential stability for $T = 5$ s, while the regular neural CLF controller results in violation of the upper bound on $V_{\theta_V}(x_t)$.

Remark 6: While we expect that our framework (see Section III) is scalable to high-dimensional systems because split CP is agnostic to the systems' dimensionality (see Lemma 5), its empirical validation is left for future work due to the page limit. Scalability can be a concern in learning prediction models [27] or safety certificates [29], but conformal robustness works independently of these processes.

V. CONCLUSION

We introduced the concept of *conformal robustness*, which provides a distribution-free, prediction-method-agnostic probabilistic bound on the model uncertainties for characterizing exponential stability and safety in fully data-driven, closed-loop, continuous-time nonlinear systems. We leveraged this concept to construct the CR-CLF and CR-CBF—explicit data-driven control designs for statistical finite-horizon guarantees of exponential stability and safety. Simulations validated our proposed concepts, showing that the CR-CLF and CR-CBF synthesized by QP or neural networks outperformed the regular, uncertainty-agnostic CLF and CBF controllers in fully data-driven settings.

REFERENCES

- [1] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York, NY, USA: Springer, 2005.
- [2] G. Shafer and V. Vovk, "A tutorial on conformal prediction," *J. Mach. Learn. Res.*, vol. 9, no. 3, pp. 371–421, 2008.
- [3] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," 2021, *arXiv:2107.07511*.
- [4] E. D. Sontag, "A Lyapunov-like characterization of asymptotic controllability," *SIAM J. Control Optim.*, vol. 21, no. 3, pp. 462–471, 1983.
- [5] Z. Artstein, "Stabilization with relaxed controls," *Nonlinear Anal., Theory, Methods Appl.*, vol. 7, no. 11, pp. 1163–1173, 1983.
- [6] R. Freeman and P. Kokotovic, *Robust Nonlinear Control Design*. Boston, MA, USA: Birkhäuser, 1996.
- [7] P. Wieland and F. Allgöwer, "Constructive safety using control barrier functions," in *Proc. 7th IFAC Symp. Nonlinear Control Syst.*, 2007, pp. 462–467.
- [8] A. D. Ames, J. W. Grizzle, and P. Tabuada, "Control barrier function based quadratic programs with application to adaptive cruise control," in *Proc. IEEE CDC*, 2014, pp. 6271–6278.
- [9] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, "Control barrier function based quadratic programs for safety critical systems," *IEEE Trans. Autom. Control*, vol. 62, no. 8, pp. 3861–3876, Aug. 2017.
- [10] M. Jankovic, "Robust control barrier functions for constrained stabilization of nonlinear systems," *Automatica*, vol. 96, pp. 359–367, Oct. 2018.
- [11] P. Florchinger, "Feedback stabilization of affine in the control stochastic differential systems by the control Lyapunov function method," in *Proc. IEEE CDC*, 1996, pp. 1047–1052.
- [12] A. Clark, "Control barrier functions for stochastic systems," *Automatica*, vol. 130, Aug. 2021, Art. no. 109688.
- [13] A. J. Taylor, V. D. Dorobantu, H. M. Le, Y. Yue, and A. D. Ames, "Episodic learning with control Lyapunov functions for uncertain robotic systems," in *Proc. IEEE/RSS IROS*, 2019, pp. 6878–6884.
- [14] A. Taylor, A. Singletary, Y. Yue, and A. Ames, "Learning for safety-critical control with control barrier functions," in *Proc. LADC*, 2020, pp. 708–717.
- [15] D. A. McAllester, "Some PAC-Bayesian theorems," in *Proc. 11th Annu. Conf. Comput. Learn. Theory*, 1998, pp. 230–234.
- [16] G. Calafiore and M. Campi, "The scenario approach to robust control design," *IEEE Trans. Autom. Control*, vol. 51, no. 5, pp. 742–753, May 2006.
- [17] L. Lindemann, Y. Zhao, X. Yu, G. J. Pappas, and J. V. Deshmukh, "Formal verification and control with conformal prediction," 2024, *arXiv:2409.00536*.
- [18] R. F. Barber, E. J. Candes, A. Ramdas, and R. J. Tibshirani, "Conformal prediction beyond exchangeability," *Ann. Statist.*, vol. 51, no. 2, pp. 816–845, 2023.
- [19] A. Dixit, L. Lindemann, S. X. Wei, M. Cleaveland, G. J. Pappas, and J. W. Burdick, "Adaptive conformal prediction for motion planning among dynamic agents," in *Proc. LADC*, 2023, pp. 300–314.
- [20] L. Lindemann, M. Cleaveland, G. Shim, and G. J. Pappas, "Safe planning in dynamic environments using conformal prediction," *IEEE Robot. Autom. Lett.*, vol. 8, no. 8, pp. 5116–5123, Aug. 2023.
- [21] J. Sun, Y. Jiang, J. Qiu, P. Nobel, M. J. Kochenderfer, and M. Schwager, "Conformal prediction for uncertainty-aware planning with diffusion dynamics model," in *Proc. NeurIPS*, vol. 36, 2023, pp. 80324–80337.
- [22] K. Y. Chee, M. A. Hsieh, and G. J. Pappas, "Uncertainty quantification for learning-based MPC using weighted conformal prediction," in *Proc. IEEE CDC*, 2023, pp. 342–349.
- [23] H. Zhou, Y. Zhang, and W. Luo, "Safety-critical control with uncertainty quantification using adaptive conformal prediction," in *Proc. ACC*, 2024, pp. 574–580.
- [24] S. Yang, G. J. Pappas, R. Mangharam, and L. Lindemann, "Safe perception-based control under stochastic sensor uncertainty using conformal prediction," in *Proc. IEEE CDC*, 2023, pp. 6072–6078.
- [25] A. D. Ames, K. Galloway, K. Sreenath, and J. W. Grizzle, "Rapidly exponentially stabilizing control Lyapunov functions and hybrid zero dynamics," *IEEE Trans. Autom. Control*, vol. 59, no. 4, pp. 876–891, Apr. 2014.
- [26] L. Weiss and E. F. Infante, "On the stability of systems defined over a finite time interval," *Proc. Nat. Acad. Sci.*, vol. 54, no. 1, pp. 44–48, 1965.
- [27] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proc. Nat. Acad. Sci.*, vol. 113, no. 15, pp. 3932–3937, 2016.
- [28] J. J. Choi, "CBF-CLF-helper 1.0: Library for control barrier function and control Lyapunov function based control methods." 2020. [Online]. Available: <https://github.com/HybridRobotics/CBF-CLF-Helper>
- [29] C. Dawson, S. Gao, and C. Fan, "Safe control with learned certificates: A survey of neural Lyapunov, barrier, and contraction methods for robotics and control," *IEEE Trans. Robot.*, vol. 39, no. 3, pp. 1749–1767, Jun. 2023.
- [30] S. Wei, P. Krishnamurthy, and F. Khorrami, "Neural Lyapunov control for nonlinear systems with unstructured uncertainties," in *Proc. ACC*, 2023, pp. 1901–1906.
- [31] Y.-C. Chang, N. Roohi, and S. Gao, "Neural Lyapunov control," in *Proc. NeurIPS*, vol. 32, 2019, pp. 1–10.